

Sanjay Suresh

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EDUCATION

University of Wisconsin–Madison

M.S. in Professional Quantum Computing (MSPQC)

Madison, WI

Starting Fall 2026

University of Wisconsin–Madison

B.S. in Computer Science and Mathematics; Physics Certificate

Madison, WI

Aug 2022 – May 2026

- **Select Coursework:** Intro to Quantum Computing, Systems Architecture for Quantum Computing, Quantum Mechanics, Cryptography, Algorithms, Analysis, Modern Algebra, Linear Optimization, Combinatorics, Operating Systems, Theory of Programming Languages, Machine Organization & Programming, Linear Algebra & Differential Equations, Discrete Mathematics

RESEARCH EXPERIENCE

Argonne National Laboratory

Quantum Computing Research Assistant — Distributed Quantum Computing

Lemont, IL

2025 – Present

- Advised by Dr. Rajkumar Kettimuthu. SeQUeNCe: **2026 R&D 100 Finalist**.
- Designed and built a distributed quantum computing module with a pluggable partitioner → scheduler → compiler stack, mapping monolithic circuits onto multi-node quantum networks under per-node memory constraints. Supports telegate, teledata, multi-hop entanglement swapping, and purification.
- Implemented a topology layer as a single source of truth for physical networks (nodes with fixed data-memory capacity, physical links, communication/EPR memory), with programmatic generators for star, grid, and connected-caveman topologies.
- Implemented multiple placement strategies behind a common interface (topology-aware capacity/distance allocation per Andrés-Martínez 2023, QAP, random, pytket-dqc), all respecting per-node memory capacity.
- Implemented scheduling strategies (ALAP, packed, serial, resource-aware) assigning each gate to a controller step and bucketing operations per node into local, remote-gate, and teleport-move roles.
- Built two compiler shapes: a static two-stage PipelineCompiler (partitioner + scheduler), and a monolithic FGPCCompiler implementing the time-sliced FGP-rOEE hybrid (Baker et al. 2020) that re-partitions qubits per time slice via relaxed OEE local search over a lookahead-weighted interaction cut.
- Instantiated the central controller as a real node in the simulated network, wired by classical channels to every processor, driving barrier-synchronized execution across nodes.
- Added a trajectory noise model: per-qubit idle T1/T2 decoherence and per-shot random Pauli errors, aggregated into success-probability statistics over many shots.
- Validated by running distributed Grover's search up to **11 qubits** and reproducing the **71% two-qubit Grover accuracy** from a published Nature DQC experiment via density matrix formalism. Faithfully replicates published DQC results (FGP-rOEE partitioning, Andrés-Martínez placement, CZ-based Grover).
- Presented at the 2025 Chicago Quantum Summit.

University of Wisconsin–Madison

Quantum Computing Research Assistant — Benchmarking (Prof. Swamit Tannu)

Madison, WI

Summer 2024 – 2025

- Developed benchmark suites with novel metrics serving as proxies for predicting resource expenditure and error rates on fault-tolerant quantum hardware.
- Leveraged variational autoencoders to investigate correlations between classically simulable metrics and actual quantum hardware performance.
- Taught a workshop on quantum computing for engineers and presented a PyEncode poster at QC 2026.

University of Wisconsin–Madison

Quantum Computing Research Assistant — Quantum Annealing (Prof. Krishnan Suresh)

Madison, WI

Summer 2023 – 2024

- Adapted quantum annealing algorithms for inequality-constrained optimization, with direct applications to real-world finance portfolio optimization.
- Enhanced algorithms for solving mixed-binary optimization problems on D-Wave quantum annealing hardware.
- Improved efficiency of solving linear systems by reducing the contraction ratio within the Box-Algorithm, achieving **30–40%** improvement on random systems of equations.
- Applied methods to 2D-Poisson, thermal, and topology optimization problems in engineering. Presented at WCCM Vancouver 2024.

University of Oxford (Remote Collaboration)

Research Collaborator — Graph State Verification (Dr. Sathyawageeswar Subramanian)

Oxford, UK

Apr 2026 – Present

- Extending graph state verification protocols to more sophisticated Pauli noise models for fault-tolerant measurement-based quantum computing.

PUBLICATIONS

- Quantum Science & Technology:** K. Suresh, S. Suresh. “PyEncode: An Open-Source Library for Structured Quantum State Preparation.”
- JCISE:** S. Suresh, K. Suresh. “Computing a Sparse Approximate Inverse on Quantum Annealing Machines.” *Journal of Computing and Information Science in Engineering*.
- Engineering Optimization:** S. Suresh, K. Suresh. “Optimal Box Contraction for Solving Linear Systems via Simulated and Quantum Annealing.” *Engineering Optimization Journal* (GENO).
- Textbook:** S. Suresh. “A Visual Approach to Programming for Middle and High School Students.” ISBN: 979-8470081841.

TALKS & PRESENTATIONS

- **QC 2026:** Workshop: Quantum Computing for Engineers. Poster: PyEncode.
- **Chicago Quantum Summit (2025):** Poster on distributed quantum computing with SeQUeNCe.
- **WCCM Vancouver (2024):** Talk on quantum annealing for linear systems and constrained optimization.

PROJECTS

Studia

Next.js, TypeScript, Convex, Vercel AI SDK, Manim, Turborepo

- Full-stack AI tutoring platform where educators create structured lessons with checkpoints and students learn through conversational AI tutoring.
- Designed a RAG pipeline that chunks uploaded PDFs, generates embeddings, and performs vector search to inject relevant course material context into AI tutoring sessions.
- Integrated Vercel AI SDK with multi-provider support (Anthropic, OpenAI) for streaming responses with real-time Manim-generated animations for visual explanations.
- Built automated review generation from student checkpoint performance. **Deployed in CS 639** (Systems Architecture for Quantum Computing, Prof. Swamit Tannu).

Eloquizzzer

Next.js, TypeScript, Convex, Anthropic, React 19, Tailwind

- Competitive quiz and learning platform with Elo-based skill tracking, adaptive difficulty, multiplayer quiz matches with real-time spectating, and personalized lesson generation across any subject.
- Science Bowl-inspired question design emphasizes reasoning and conceptual transfer over recall.

SurfaceScope (In Progress)

Python, Flask, React, TypeScript, Stim, PyMatching

- Interactive web-based visualizer for surface-code quantum error correction, enabling step-by-step exploration of noise injection, syndrome extraction, detector error models, and minimum-weight perfect matching decoding.
- Implemented a multi-qubit pipeline that compiles OpenQASM circuits to native Clifford gates, routes qubits with SWAP insertion on a linear array, and simulates per-patch surface code error correction with Stim.
- Built threshold estimation tooling running batch simulations across code distances and physical error rates, producing log-log plots of logical vs. physical error rates.
- Designed interactive SVG lattice rendering with pan/zoom, tooltips, manual error injection, and animated pipeline playback across six decoding stages.

PDFQuizzzer

Next.js, TypeScript, Convex, PDF.js, KaTeX

- Interactive study tool that generates quizzes from uploaded PDF documents. Upload, read, quiz page by page, and review performance with streak tracking and KaTeX math rendering.

AIManager

Next.js, TypeScript, Convex, Vercel AI SDK, Anthropic, OpenAI

- Personal productivity platform with AI-powered task management, calendar integration, habit tracking, motivational dashboard, pinboard, and knowledge base with integrated chat assistant.

LEADERSHIP & TEACHING

Printable Programming

Madison, WI

Principal Instructor

2019 – Present

- Authored *A Visual Approach to Programming for Middle and High School Students* (ISBN: 979-8470081841) — a comprehensive textbook for CAD modeling, programming, and 3D-printing.
- Instructed underprivileged students at the Boys and Girls Club of Dane County in STEM education.
- Contributed to the development of the CADjs software application.

Science Bowl

Middleton, WI

President & Captain

2020 – 2022

- Ranked **2nd nationally out of 700+ individuals** based on overall score.
- Founded Summit Scibowl — a for-profit dedicated to advanced training of students in Science Bowl.
- Built online platforms engaging 2,000+ students nationwide and organized multiple national tournaments.

HONORS & AWARDS

- **Atlas Fellowship (2022):** Selected from **7,000+** applicants worldwide (**114 recipients**) for a \$50K fellowship recognizing exceptional aptitude in STEM. Attended the Atlas Fellowship Summer Program.
- **AIME Qualifier:** American Invitational Mathematics Examination.
- **Chemistry Olympiad:** National Qualifier (2021 & 2022).
- **National Astrophysics Olympiad:** Qualifier (2021).
- **Earth Science Olympiad:** Training Camp Qualifier (2022).
- **Wisconsin Math Talent Search:** Finalist (2022).
- **Mathematics Student of the Year:** Middleton High School (2022).
- **Naval Research Award:** Capital Science and Engineering Fair (2020).
- **Prometheus Science Bowl:** Individual 2nd Place.

SKILLS

Languages: Python, TypeScript, JavaScript, Java, C, Haskell, MATLAB

Quantum Frameworks: Qiskit, PennyLane, D-Wave Ocean, Qutip, Stim, PyMatching

Web & Infrastructure: Next.js, React, Convex, Vercel AI SDK, Flask, Turborepo, Tailwind CSS, PostgreSQL, Docker, Git

AI & ML: Vercel AI SDK, LangChain, LangGraph, OpenAI API, Anthropic API, RAG pipelines

Testing: Jest, Pytest

OTHER

Carnatic Violin: Professional violinist and accompanist to frontline artists at major festivals across India and the US. Trained since 2012; performing as soloist and accompanist since 2017. Has accompanied artists including Smt Alamelu Mani, Shri Sikkil Gurucharan, Akkarai Sisters, Dr. S Sowmya, Carnatica Brothers, and Dr. Gayathri Girish, sharing the stage with legends such as Shri Trichy Sankaran and Shri Patri Sathish Kumar.